Project Documentation: Autonomous B2B Lead Workflow

1. Project Overview ■

This project is an end-to-end multi-agent system designed to automate the B2B outbound lead generation process. It uses the **LangGraph** framework to create a workflow of autonomous agents that work collaboratively to discover, enrich, score, and contact B2B prospects. The entire system is dynamically configured from a single **workflow.json** file, allowing for easy modification and extension. A key feature is the **FeedbackTrainerAgent**, which analyzes campaign results and suggests improvements, creating a continuous learning loop.

2. Core Features

• Dynamic Workflow Engine: Defined in workflow.json, dynamically builds agent graphs.
• Modular Agent Architecture: Self-contained classes for reusability.
• End-to-End Automation: Covers full lead generation cycle.
• AI-Powered Content Generation: Uses GPT-4o-mini for personalized messaging.
• Adaptive Feedback Loop: Logs results to Google Sheets and suggests improvements.
• Secure Configuration: Sensitive info stored in .env file.

3. Technology Stack ■■

Component	Tool / Library	Purpose
Agent Framework	LangGraph & LangChain	Core orchestration and state management
LLM	OpenAI GPT-4o-mini	Reasoning and personalized content generation
Data APIs	Clay API, Apollo API	Prospect and company data retrieval
Enrichment	Clearbit / PeopleDataLabs	Augmenting lead data with valuable context
Email Delivery	SendGrid API	Executing the email outreach campaigns
Feedback Logging	Google Sheets API	Storing campaign results and recommendations

4. Setup and Installation ■

Follow these steps to set up and run the project locally. **Step 1:** Ensure you have Python 3.10 or newer installed.
Step 2: Create and activate a virtual environment:

```
python -m venv venv
source venv/bin/activate (macOS/Linux)
venv\Scripts\activate (Windows)
```